



Connor J. Stone PhD

ASTROPHYSICIST · OPEN SOURCE DEVELOPER · SCIENCE COMMUNICATOR

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"I do Astrostatistics and software development for supernovae, astronomical images, and gravitational lensing"

Current Positions

Postdoctoral Fellow University of Toronto, Dunlap Institute (Sept. 2025 - Exp. Aug. 2028)
Rubin Fellow Rubin Observatory, via University of Toronto (Sept. 2025 - Exp. Aug. 2028)

Previous Positions

Postdoctoral Fellow Université de Montréal (Sept. 2022 - Exp. Aug. 2025)
Joint Postdoctoral Fellow Mila (Sept. 2022 - Exp. Aug. 2025)
CITA National Fellow Canadian Institute for Theoretical Astrophysics (Sept. 2022 - Exp. Aug. 2025)
NSERC Postdoctoral Fellow Natural Sciences and Engineering Research Council of Canada (Nov. 2022 - Exp. Aug. 2024)
Queen's Observatory Coordinator Queen's University (June. 2018 - Apr. 2022)

Education

Queen's University (Department of Physics, Engineering Physics & Astronomy)

Ph.D. Astrophysics

- **Topic:** Deciphering the Complexity of Galaxy Structure: New Catalogs, Deep Photometry, and Intrinsic Scatter
- **Advisor:** Professor Stéphane Courteau

Kingston, Ontario

May. 2018 - Aug. 2022

Queen's University (Department of Physics, Engineering Physics & Astronomy)

M.Sc. Physics

- **Topic:** From Surface Events in DEAP-3600 to the Radial Acceleration Relation in Spiral Galaxies
- **Advisors:** Professor Anthony Noble and Professor Stéphane Courteau

Kingston, Ontario

May. 2016 - Aug. 2018

University of Waterloo (Department of Physics & Astronomy)

B.Sc. Physics

- Co-operative program
- Dean's Honours List

Waterloo, Ontario

Sept. 2011 - Apr. 2016

Publications

COMPLETE RECORDS

First author publications

<https://scixplorer.org/public-libraries/855ilJitQxKOS5q1XQh17g>

NASA ADS

All (co-)authored publications

<https://scixplorer.org/public-libraries/VX2TrCdNQ42VR0Y6xpu6kQ>

SciXplorer

HIGHLIGHTED CONTRIBUTIONS

[6] Pixellated Posterior Sampling of Point Spread Functions in Astronomical Images

Connor Stone; Ronan Legin; Alexandre Adam; Esmeralda Whitammer; Gabriel Missael Barco; Laurence Perrault-Levasseur; Yashar Hezaveh.

ApJ (sub.)

June 2026

[5] Mind the Information Gap: Unveiling Detailed Morphologies of z 0.5-1.0 Galaxies with SLACS Strong Lenses and Data-Driven Analysis

Ronan Legin; Connor Stone; Alexandre Adam; Gabriel Missael Barco; Adam Coogan; Esmeralda Whitammer; Laurence Perrault-Levasseur; Yashar Hezaveh.

ApJ (sub.)

June 2026

- [4] **Caustics: A Python Package for Accelerated Strong Gravitational Lensing Simulations** *JOSS*
Connor Stone; Alexandre Adam; Adam Coogan; M. J. Yantovski-Barth; Andreas Filipp; Landung Setiawan; Cordero Core;
 Ronan Legin; Charles Wilson; Gabriel Missael Barco; Yashar Hezaveh; Laurence Perrault-Levasseur. *June 2024*
- [3] **Echoes in the Noise: Posterior Samples of Faint Galaxy Surface Brightness Profiles with Score-Based Likelihoods and Priors** *AJ*
 Alexandre Adam; **Connor Stone**; Connor Bottrell; Ronan Legin; Yashar Hezaveh; Laurence Perrault-Levasseur. *Nov. 2023*
- [2] **AstroPhot: Fitting Everything Everywhere All at Once in Astronomical Images** *MNRAS*
Connor Stone; Stéphane Courteau; Jean-Charles Cuillandre; Yashar Hezaveh; Laurence Perrault-Levasseur; Nikhil Arora. *Aug. 2023*
- [1] **The Intrinsic Scatter of Galaxy Scaling Relations** *ApJ*
Connor Stone; Stéphane Courteau; Nikhil Arora. *ApJ 912: 41* *May 2021*

Grants, Scholarships, & Awards

NOVA FRQNT-NSERC	<i>FRQNT and NSERC</i>
Amount: \$210,000 over 3 years; Competitive, jointly awarded with Co-Is at 3 other institutions	<i>Mar. 2023 - Feb. 2026</i>
NSERC Postdoctoral Fellowship	<i>Natural Sciences and Engineering Research Council of Canada</i>
Amount: \$90,000 over 2 years; Competitive	<i>Nov. 2022 - Aug. 2024</i>
CITA National Fellowship	<i>Canadian Institute for Theoretical Astrophysics</i>
Amount: \$105,000 over 3 years; Competitive	<i>Sept. 2022 - Aug. 2025</i>
The Walter C. Sumner Memorial Fellowship	<i>Walter C Sumner Foundation</i>
Amount: \$8,000; Competitive	<i>Sept. 2021 - Apr. 2022</i>
Natural Sciences and Engineering Research Council of Canada (NSERC) Postgraduate Scholarship – Doctoral (PGS-D)	<i>Government of Canada</i>
Amount: \$63,000 over 3 years; Competitive	<i>Sept. 2019 - Aug. 2022</i>
Arthur B. McDonald Prize for Academic Excellence	<i>McDonald Institute</i>
Amount: \$30,000; Competitive	<i>Sept. 2018 - Aug. 2019</i>
Ontario Graduate Scholarship	<i>Province of Ontario</i>
Amount: \$30,000 over 2 years; Competitive	<i>Sept. 2016 - Aug. 2018</i>
Graduate Entrance Tuition Award	<i>Queen's University</i>
Amount: \$6,414; Grade Based	<i>May 2016</i>
Queen Elizabeth II Aiming for the Top Scholarship	<i>Province of Ontario</i>
Amount: \$14,000 over 5 years; Grade Based	<i>Sept. 2011 - Apr. 2016</i>
President's Scholarship	<i>University of Waterloo</i>
Amount: \$2,000; Grade Based	<i>Sept. 2011</i>
Physics Alumni Award	<i>University of Waterloo</i>
Amount: \$1,000; Grade Based	<i>Sept. 2011</i>

Teaching & Supervision

DIRECT SUPERVISION

Students I have directly supervised. Guiding their project and acting as a first contact for the research process.

Zoe Augspach	<i>Undergraduate</i>
Project: Forward modeling AGN light curves in Rubin.	<i>May 2026 - Present</i>
Sulaiman Alangari	<i>Undergraduate</i>
Project: Inferring the SNIa SED from Rubin photometry.	<i>May 2026 - Aug. 2026</i>

Linh Vo	<i>Undergraduate</i>
Project: Inference method validation for cosmological dark energy equation of state.	May 2026 – Aug. 2026
Sacha Perry-Fagant	<i>PhD</i>
Project: Posterior sampling of Star Formation Histories in Galaxies.	Sept. 2023 – Aug. 2025
Raj Patel	<i>MSc</i>
Project: The Universal Rotation Curve of Disk Galaxies.	Sept. 2021 – Nov. 2023
Brianna Nix	<i>Undergraduate</i>
Project: studying the halos of Virgo Cluster galaxies.	Sept. 2021 – Apr. 2022
Jacob Meadus	<i>Undergraduate</i>
Project: studied SED fits on reduced flux measurement data for PROBES galaxies.	Sept. 2020 – Apr. 2021
Katherine Fraser	<i>Undergraduate</i>
Project: studied PROBES galaxies with complex morphologies, extraction of WISE photometry.	May 2020 - Aug. 2020
Maggie Oxford	<i>Undergraduate</i>
Project: studied scaling relations in super-massive black hole hosts.	May 2020 - Aug. 2020
Kelsey Eakin	<i>Undergraduate</i>
Project: studied inner rotation curve slopes versus photometric properties of PROBES galaxies.	May 2019 – Apr. 2020

MENTOR/COLLABORATOR SUPERVISION

Students for whom I was academically senior while collaborating and/or acting as a mentor.

Callista Sullivan	<i>MSc</i>
Project: Simulation Based Inference for modeling gaps and rings in planetary disks	Sept. 2025 – Present
Lawrence Faria	<i>PhD</i>
Project: Inferring high resolution rotation curves from HI velocity cubes using Diffusion models	Sept. 2024 – Present
Gabriel Missael-Barco	<i>PhD</i>
Project: Tackling the Problem of Distributional Shifts: Correcting Misspecified, High-dimensional Data-driven Priors for Inverse Problems	Sept. 2023 – Aug. 2025
Ronan Legin	<i>PhD</i>
Project: Mind the Information Gap: Unveiling Detailed Morphologies of z 0.5-1.0 Galaxies with SLACS Strong Lenses and Data-Driven Analysis	Sept. 2023 – Aug. 2025
Alexandre Adam	<i>PhD</i>
Project: Echoes in the Noise: Posterior Samples of Faint Galaxy Surface Brightness Profiles with Score-based Likelihoods and Priors	Sept. 2023 – Aug. 2025
Zahra Basti	<i>MSc</i>
Project: Evaluating mathematical models for galaxy mass distribution	Sept. 2021 – March 2023
Sean Begy	<i>MSc</i>
Project: Angular momentum tests for dark matter in galaxies	Sept. 2018 – Aug. 2021
Matt Frosst	<i>MSc</i>
Project: The diversity of spiral galaxies explained	Sept. 2018 – Aug. 2022

Presentations_____

Aug. 2026	Building a reliable simulator at scale , Conference talk	JSM Boston
July 2026	Caustics: a differentiable, GPU accelerated, gravitational lensing simulator , Conference talk	RCW California
July 2026	AstroPhot: Fitting Every-transient Everywhere all at Once , Conference talk	RCW California
July 2026	AstroPhot: Fitting Everything Everywhere all at Once , Conference talk	NAM Birmingham
July 2026	Building Forward Models for Astronomical Applications in the caskade Ecosystem , Workshop	STATSTRO UofT
June 2026	Practically perfect pixelated PSF posterior samples , Poster	Université de Montréal
June 2026	Simulating transients for the next generation of surveys , Poster	Université de Montréal
March 2026	Making scientific simulators for inference problems , Team talk	University of Toronto
Jan. 2026	Rethinking supernova science from the ground up, with scene modelling and Bayesian Cosmology , Conference talk	AAS, Phoenix
Oct. 2025	Two sample tests and Bayesian PSFs , Team talk	University of Toronto
Oct. 2025	CanDIAPL workshop: Rubin Supernova Dark Energy Science , Workshop talk	University of Waterloo
Sept. 2025	AstroPhot: Fitting Every Supernova Everywhere All at Once , Collaboration talk	LSST DESC
July 2025	AstroPhot: Fitting Everything Everywhere All at Once in Astronomical Images , Workshop talk	University of Manitoba
July 2025	Caustics: a differentiable, GPU accelerated, gravitational lensing simulator , Workshop talk	Tucson, AZ
June 2025	CASCADE: making scientific simulators and pipelines easy , Conference talk	St. Mary's University
May 2025	World Coordinate Systems in Astronomy , Workshop talk	Ciela Institute
May 2025	Pixel level Bayesian analysis of PSFs and Strong Gravitational Lenses , Conference talk	Paris
May 2025	CASCADE: building scientific simulators, solving the args problem , Collaboration talk	LSST DESC
May 2025	Pushing the limits of Bayesian analysis for gravitational lensing , Invited talk	American Museum of Natural History
Sept. 2024	Easy A: Getting a Perfect Score in Astronomical Image Processing , Invited talk	Queen's University
Aug. 2024	Ready Player One: Getting a Good Score in Observational Astronomy , Invited talk	Universe TBD
Aug. 2024	Caustics: a differentiable, GPU accelerated, gravitational lensing simulator , Conference talk	South Africa
Aug. 2024	Linear Regression: a Machine Learning Perspective , Invited Lecture	Kingston, Ontario
July 2024	Understanding Bayesian Statistics with the On/Off Problem , Invited Lecture	Kingston, Ontario
July 2024	Development of AstroPhot and Caustics: Fitting Everything Everywhere all at Once , Conference talk	Tacoma, USA
June 2024	Bayesian Statistics and Linear Regression: a Machine Learning Perspective , CRAQ summer school Lecture	Montréal, Québec
June 2024	Caustics: the gravitational lensing simulator of the future , Conference talk	Toronto, Ontario
May 2024	Caustics: A Differentiable, GPU Accelerated, Gravitational Lensing Simulator , Conference talk	Rome, Italy
March 2024	Caustics: The gravitational lensing simulator of the future , Contributed tutorial session	Oxford, UK
Jan. 2024	Caustics: The gravitational lensing simulator of the future , Poster	New Orleans, USA
Oct. 2023	AstroPhot: Fitting Everything Everywhere All at Once in Astronomical Images , Conference talk	Banff, Alberta
Aug. 2023	AstroPhot: Fitting Everything Everywhere All at Once in Astronomical Images , Seminar	Waterloo, Ontario
April 2023	Ready Player One: Getting a Good Score in Observational Astronomy , CITA Seminar	Toronto, Ontario
May 2023	Surveys 2 Discoveries , tutorial session	Montréal, Quebec
Jan 2023	Timescales in Astrophysics , Conference talk	United Arab Emirates
May 2022	Canadian Astronomy in the Roaring 2020's: CASCA 2022 , Poster and flash talk	Virtual
2017-2021	Queen's university, Department of Physics, Engineering Physics & Astronomy , Twice yearly Journals Club	Kingston, Ontario
May 2021	Canadian Astronomical Society, Société Canadienne Astronomie 2021 , Poster and flash talk	Virtual
July 2020	Arthur B. McDonald Institute, Tools from the Couch , Seminar	Virtual
June 2020	Arthur B. McDonald Institute, Tools from the Couch , Seminar	Virtual
Aug 2019	Santa Cruz Workshop on Galaxy Formation , Poster and flash talk	Santa Cruz, CA
July 2018	The Physics of Galaxy Scaling Relations and the Nature of Dark Matter , Poster and flash talk	Kingston, Ontario
May 2017	The Canadian Association of Physicists Congress , Talk	Kingston, Ontario
June 2015	The Canadian Association of Physicists Congress , Poster	Edmonton, Alberta

Public Outreach

PRESS/MEDIA APPEARANCES

Jan. 2021	Grad Chat Podcast , Queen's School of Graduate Studies	Kingston
Dec. 2020	Celestial Occurrences in December 2020 , Global News	Kingston
Sept. 2020	PhD candidate Connor Stone explains the discovery of phosphine in the clouds of Venus , Discovery Canada	Virtual
July 2020	Comet remains to be seen for the month of July , Whig Standard	Kingston
June 2020	Queen's University astrophysicist sheds some light on the historic SpaceX launch , Global News	Kingston
June 2020	First Crewed SpaceX Launch , CFRC	Kingston
Apr. 2020	Spotting Comet ATLAS C/2019 Y4 , YourTV Kingston	Kingston
Nov. 2019	Mercury moves across the sun , Global News	Kingston
July 2019	Canada's contributions to the Apollo 11 mission , Global News	Kingston
June 2019	A Nobel Pursuit , The Queen's Gazette	Kingston
Dec. 2018	Queen's Phd student tells us how we can spot the International space station in the sky , Global News	Kingston

PUBLIC TALKS

Feb. 2026	Supernovae: tick marks in the cosmic tape measure , Astronomy on Tap	Kingston
March 2023	Mysterious Galaxies at the Beginning of Time , Queen's Observatory	Kingston
May 2022	Virtual Planetarium Keynote , Science Rendezvous Kingston	Virtual
May 2021	Virtual Planetarium Keynote , Science Rendezvous Kingston	Virtual
June 2019	Where Does it All Come From? , Queen's Observatory	Kingston
May 2019	Young Persons Lecture Competition (Honourable Mention) , Institute of Materials, Minerals, & Mining	Kingston
Oct. 2018	Grad Student Panel Discussion: Get the Real Deal , School of Graduate Studies	Kingston
Oct. 2018	Tools for Machine Learning , Queen's Machine Intelligence and Neuroevolution Design (Q-MIND)	Kingston
Feb. 2018	What is Dark Matter? , Queen's Observatory	Kingston

Service

Pixels-to-Light-Curves co-Convener

Coordinating activities in DESC related to the processing of images into calibrated transient light curves	LSST DESC
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Astronomy Data Analysis Software & Systems 2027	University of Toronto
Member of the LOC. Lead for conference proceedings, help with scheduling and design materials	Oct. 2025 - Oct. 2027

Rubin ISSC DP2 Hackathon	Carnegie Mellon University
Member of the SOC. Lead for hackathon project, some event organization	Sept. 2026 - Jan. 2027

Ciela Institute Professional Development Series	Université de Montréal
Tutorials on professional applications of tools like Blender, GIMP, Audacity, etc.	July 2023 - Aug. 2025

Ciela Institute Astromatic Hackathon	Université de Montréal
Lecture, plus LOC responsibilities for the week long hackathon	August 2023

UNIONS data working group meeting	Université de Montréal
LOC and primary event coordinator	May 2023

Surveys 2 Discoveries conference	Université de Montréal
Member of the LOC. Organizing payment system, booking logistics, forms and attendee data.	May 2023

UdeM Astronomy Department Seminar Committee	Université de Montréal
Organizing and coordinating seminar program including inviting international speakers to the department.	Sept. 2022 - Aug. 2025

Equity, Diversity, and Inclusion Committee	Ciela Institute
Organizing and coordinating EDI activities in the Institute including the development of the code of conduct.	Sept. 2022 - Aug. 2025

Journal Referee	ApJL, ApJ, JOSS, ICML, CanTAC
The ApJL, ApJ, JOSS, ICML, and CanTAC	2021 - Present

Physics Department Outreach Committee	Queen's University
Organizing and coordinating outreach activities in the Physics department.	Sept. 2021 - Aug. 2022

Q-GraPh Graduate Physics Journals Club	Queen's University
Creator/Organizer. Year-round weekly Journals Club. Organized over 170 talks.	May 2017 - Apr. 2022

Data Science Discussion Group

Co-Creator/Organizer. Year-round monthly discussions on Machine learning and Data Science

The Physics of Galaxy Scaling Relations and the Nature of Dark Matter

Member of the LOC, involved in various elements of conference organization and operations.

Queen's University

Sept. 2017 - Sept. 2019

Queen's University

July 2018